

Ad Tag Tracking System — Non-Technical Specification & Build Guide

Purpose: Monitor selected websites for digital advertising tags, track changes over time, and deliver a monthly report and notifications. This guide is written in plain language for a newcomer to software and product development.

1) Product Overview

- **What it does:** Scans chosen websites on a schedule, detects advertising tags, keeps a history of changes, and notifies people. It shows activity in dashboards and sends monthly usage reports.
 - **Who uses it:**
 - **User:** Can add and manage Sites, view scan results and reports.
 - **Admin:** Everything a User can do plus user management, system settings, and data exports.
 - **Key outcomes:** 1) Clear visibility into which advertising tags are present per Site 2) Reliable change detection with a historical timeline 3) Monthly summaries for both **usage** and **tag changes**
-

2) Definitions

- **Site:** A brand/client domain (e.g., `brand.com`) to be monitored.
 - **Advertising tag:** Any script, pixel, or URL that belongs to an ad/measurement platform (e.g., Google Ads, Meta Pixel, Trade Desk, etc.).
 - **Scan:** A visit to a Site that collects the page HTML, network requests, and tag evidence, using selected geolocation and device type.
 - **Change:** A difference in tag presence or tag parameters compared with the previous scan.
-

3) User Roles & Permissions

- **User**
 - Create, edit, pause (inactivate), and delete their Sites
 - View dashboards, scan history, tag details, and reports
 - Set notifications (email, ClickUp)
 - **Admin** (includes all User permissions +)
 - Create/manage users (reset passwords, assign roles)
 - Manage system-wide tag domain lists (sources + manual entries)
 - Configure default scan settings and quotas
 - Access and export usage logs
-

4) Security & Account Basics

- **Secure login** with email + password
 - **Password reset** via email link (token expires)
 - **Multi-factor authentication (MFA) (recommended)**
 - **Role-based access control** (User, Admin)
 - **Session security** (short tokens, inactivity timeouts, revoke on password change)
 - **Encryption** in transit (HTTPS) and at rest for sensitive data
 - **Audit logs**: logins, settings edits, and admin actions
-

5) Inputs (Per Site)

For each Site, allow the following editable fields: 1. **Domain(s)** to scan (primary + optional subpages list or sitemap URL) 2. **Scan Frequency** (e.g., daily, every 3 days, weekly) — configurable per Site 3. **Status**: Active / Inactive 4. **End Date** for scanning (optional) 5. **Geo Locations** to scan from: country + state/province (multi-select) 6. **Device Type**: Desktop, Mobile, or Both (controls the user agent) 7. **Email(s)** for change notifications 8. **ClickUp email** for automatic task creation. Task tagged “**Bus. Dev.**” and a notification sent to the Bus Dev Team. Include details of the change (which tag, when, where).

6) Inputs (System-Wide)

1. **Known tag domain sources** (e.g., links to curated lists)
 2. **Manual tag domains** (admin can add/edit/remove)
 3. **User management** (create users, assign role, password reset)
 4. **Quick link** to **Create New Site**
-

7) Dashboards & Views

7.1 Main Dashboard (post-login)

- **Sites table** (sortable, filterable):
 - Site domain
 - Status (Active/Inactive)
 - Last scan status (Success/Failed + short error if failed)
 - Last scan date/time
 - “Changes since last scan” (Yes/No + count)
- **Create New Site** button/section with short helper text
- **Charts** (examples):
 - Scans completed over time (last 30/90 days)
 - Tag changes detected over time
 - Success vs. failed scans

7.2 Site Dashboard

- **Header:** Site domain + thumbnail/screenshot of homepage
- **Status & Scan Info:** Active/Inactive, Last scan status, Last scan date/time, Changes since last scan
- **Tag Changes feed:**
 - Which advertising tag changed (name/domain)
 - Change date/time
 - Was it seen before? (Yes/No; show first seen date)
- **Tag Details (link from each tag):**
 - Full tag URL or script source
 - Any identifiable IDs in the tag (e.g., account IDs)
 - Owner or associated company (if known)
 - History timeline (first seen, last seen, diffs)
- **Editable Settings** (from Section 5) for this Site

7.3 Create New Site

- Simple form collecting all **Per Site Inputs** (Section 5)
- Validation (e.g., domain format, at least one frequency & geolocation)
- Test scan button (optional) to verify basic access

7.4 Usage Reports (on-screen)

- Date filters (e.g., last month)
- Show:
 - User logins (who, when)
 - Session duration (rough estimate of “time for each login”)
 - Site scan dates and counts
 - Notifications sent (email + ClickUp)
 - Export as CSV and/or PDF

8) Notifications & Monthly Emails

- **Change Notifications (near-real-time):**
 - Email to the provided addresses
 - ClickUp email creates/updates a task in the target List. Include: Site, page URL, detected tag, what changed, when, and attach a diff or short summary. Apply “**Bus. Dev.**” tag and notify the Bus Dev Team.
 - **Monthly Usage Logs (auto-email):**
 - Sent to configured recipients
 - Includes user login counts, average session time, number of scans, failures, and number of notifications sent
 - **Monthly Tag Summary (auto-email):**
 - Sites scanned, new tags found, removed tags, changed tags
 - Top Sites by volume of changes
-

9) Scanning & Detection (Conceptual)

- **How a scan works (simple view):** 1) Use a headless browser to open the Site (desktop or mobile user agent) 2) Load the homepage (and optionally key subpages/sitemap) 3) Collect:
 - Final HTML
 - Network requests (URLs called during page load)
 - Detected scripts/pixels that match the known tag domains4) Store the results and compare with the previous scan for that Site
 - **Geolocation:** Use proxy IPs for selected countries/states/provinces
 - **Timing:** Wait for the page to settle (e.g., DOM loaded + a short extra delay) to catch late-loading tags
 - **Change detection:**
 - Compare sets of tag domains and key parameters (e.g., account IDs) between scans
 - Classify as **Added**, **Removed**, or **Modified**
 - **Error handling:**
 - Capture clear errors (timeout, blocked by bot protection, DNS error, etc.)
 - Retry once if safe; otherwise record failure with details
 - **Evidence:**
 - Keep a minimal snapshot (DOM hash, network log excerpt, and optional screenshot) for the audit trail
-

10) Data Model (Plain-English)

- **User:** name, email, role, password hash, MFA status, created/updated dates
 - **Site:** domain, status, frequency, end date, geos, device setting, emails, ClickUp email, created/updated dates
 - **Scan:** site, datetime, geo, device, status, error (if any), raw artifacts (HTML hash, network log link), counts
 - **Tag:** normalized name (e.g., "Google Ads"), matching rule(s) (domains/patterns), company link
 - **Detection:** scan, tag, URL, parameters (e.g., IDs), first seen date, last seen date
 - **Change Event:** site, when, tag, change type (Added/Removed/Modified), details, linked old/new values
 - **Usage Log:** user, login time, logout time (or last active), actions
 - **Notification:** type (email/ClickUp), recipients, subject, body summary, send status, related change event(s)
-

11) Charts to Include

- Scans by day/week
 - Changes by day/week
 - Failure rate over time
 - Top tags detected (count)
 - Top Sites by number of changes
-

12) Error Checking & Reporting

- **Validation:**
 - Domain format, unique Site per domain, at least one geo and frequency
 - Emails are valid
 - **Operational errors:**
 - Clear messages on dashboard (e.g., "Timeout fetching homepage")
 - Admin error log view with filters
 - **Auto reporting:**
 - Email summary of repeated failures (e.g., Site failing >3 scans in a row)
-

13) Privacy, Security, and Compliance Basics (Practical)

- Avoid storing full page HTML unless necessary (store a hash + small excerpt)
 - Avoid storing personally identifiable information (PII) from pages
 - Respect robots.txt where appropriate; if you must scan, add a configurable policy toggle
 - Protect API keys and proxy credentials (secure vault)
 - Rate limit scans and respect target Sites (no overwhelming traffic)
-

14) Monthly Reports (Content Template)

Subject: Monthly Tag & Usage Summary — {{Month YYYY}}

Body (high-level): 1) **Usage** - Users active: X - Logins: X (avg session Y mins) - Total scans: X (success: A, failed: B) - Notifications sent: Email C, ClickUp D 2) **Tag Changes** - New tags found: X (top 5) - Removed tags: Y (top 5) - Modified tags: Z (examples) 3) **Top Sites by Change Volume** - 1) domain — N changes - 2) domain — M changes

Attach CSVs (changes.csv, usage.csv) and link to the online report page.

15) Acceptance Criteria (Checklists)

A. Authentication & Roles

- ☐ Users can sign up/in securely and reset password
- ☐ Admin can create users and set User/Admin roles
- ☐ Sessions expire on inactivity and can be revoked

B. Site Management

- ☐ Create/Edit a Site with all fields in Section 5
- ☐ Set per-Site scan frequency

- ☐ Pause (Inactive) and resume
- ☐ End date stops further scans

C. Scanning

- ☐ Runs on the chosen schedule for each Site
- ☐ Respects geo/device settings using proxies and user agents
- ☐ Captures network calls and detects tags from known lists + manual entries
- ☐ Records success/failure with clear reasons

D. Change Detection & History

- ☐ Classifies Added/Removed/Modified
- ☐ Shows first seen / last seen dates
- ☐ Provides tag details (URL, IDs, owner link)

E. Notifications

- ☐ Email sent on change with clear summary + deep link
- ☐ ClickUp task created via email with **"Bus. Dev."** tag and team notified

F. Dashboards & Reports

- ☐ Main dashboard shows Sites with fields in 7.1
- ☐ Site dashboard shows details in 7.2
- ☐ Usage reports show items in 7.4 and export
- ☐ Monthly usage logs are emailed to configured recipients
- ☐ Monthly tag summary email is sent

G. Reliability & Safety

- ☐ Retries where safe; avoids infinite loops
- ☐ Error logs viewable by Admin; recurring issues summarized
- ☐ Reasonable rate limits to avoid overloading target Sites

16) Suggested Build Steps (Plain, New-Friendly)

1. **Define success:** Pick 3–5 Sites to monitor in a pilot.
2. **UX design (lo-fi first):** Sketch the Main & Site dashboards and the Create Site form (paper or a tool like Figma). Keep it simple.
3. **Choose a build path** (see Options below). For a quick start, pick one stack and stick to it.
4. **Set up your workspace:** Create a repo (Git), choose a cloud dev environment (Replit or alternatives), and add a README.
5. **Start small:** Build authentication + the "Create Site" form and listing page.
6. **Implement scanning** for a single Site, one geo, one device. Confirm it detects at least one known tag.
7. **Add change detection:** Compare two scans and show Added/Removed/Modified.

8. **Wire notifications:** Email first; then ClickUp via email.
 9. **Extend:** Multiple geos/devices, better error handling, charts, and exports.
 10. **Polish reports:** Monthly emails + downloadable CSV.
 11. **Test** end-to-end with the pilot Sites. Fix rough edges.
 12. **Document** how to add a new Site and how to read results.
-

17) Tech Options (Pick One Path)

Option A — JavaScript/TypeScript (full-stack)

- **Frontend:** React (or Next.js) for dashboards and forms
- **Backend:** Node.js + Express (or Next.js API routes)
- **Scanning:** Playwright or Puppeteer (headless browser)
- **Database:** Postgres (or SQLite to start)
- **Background jobs:** Simple cron (or hosted scheduler)
- **Email:** Any transactional email service (e.g., SendGrid)
- **Pros:** One language end-to-end; strong ecosystem; easy charts

Option B — Python (pragmatic)

- **Frontend:** Simple React or server-rendered templates
- **Backend:** FastAPI (or Django)
- **Scanning:** Playwright for Python
- **Database:** Postgres/SQLite
- **Background jobs:** Celery/cron
- **Pros:** Clean syntax, lots of libraries, great for scripting and data

Option C — Low-Code + Scripts (fastest to MVP)

- **Frontend:** Retool/Tooljet (admin UIs) + a simple public viewer
 - **Backend:** Small API for scans + a task runner
 - **Scanning:** Playwright/Puppeteer scripts
 - **Database:** Hosted Postgres
 - **Pros:** Speed to first usable version; visual tables/forms
-

18) Cloud Dev Environments (AI-Friendly)

- **Replit** (AI coding built-in, easy start)
 - **Alternatives:** GitHub Codespaces (VS Code in the cloud), StackBlitz, CodeSandbox, Glitch. Pick one; they all let you code from the browser and share.
-

19) Integration Notes (ClickUp & Email)

- **ClickUp (email-to-task):** Use the ClickUp List's unique email address. Subject can include the Site and "Tag Change". In the body, include:
 - Site domain + page URL
 - Tag name and linked company
 - Change type with brief before/after
 - Date/time (UTC)
 - Apply "**Bus. Dev.**" tag
 - Mention/notify Bus Dev Team (per your ClickUp setup)
 - **Email notifications:** Keep them short with a deep link back to the Site dashboard and the specific Change Event.
-

20) Simple UI Copy (Examples)

- **Create Site Help Text:** "We'll scan this domain on a schedule you choose. Pick where we scan from (country/state) and whether we mimic desktop, mobile, or both."
 - **Empty State (No Tags):** "No advertising tags detected yet. We'll notify you if anything shows up."
 - **Error Banner:** "Last scan failed (Timeout). We'll retry on the next scheduled run. See error details."
-

21) Testing Plan (New-Friendly)

- **Unit checks:** Does URL validation work? Do we store selections correctly?
 - **Scan sandbox:** Point at a known test page with a sample tag. Confirm detection.
 - **Diff checks:** Add/remove a test tag and confirm Added/Removed/Modified shows correctly.
 - **Notification dry-run:** Send emails to yourself and verify ClickUp creates a task.
 - **Permissions:** Log in as User vs Admin and confirm differences.
-

22) What to Build First (MVP Scope)

1) Auth + Roles 2) Create Site (single domain, one geo, desktop only) 3) Manual "Run Scan Now" + nightly schedule 4) Detect from a small **manual tag domain list** 5) Change history on the Site page 6) Email on change (no ClickUp yet)

Ship this, then expand.

23) Educational Links (Beginner-Friendly)

See the companion message for direct links. - UX basics & wireframing - Git & GitHub fundamentals - Web scraping/headless browsers (Playwright/Puppeteer) - Product delivery basics (Agile) - Security basics (web app hygiene)

24) Future Enhancements (Ideas)

- Crawl selected subpages (menu, sitemap) with depth limits
- Detect parameter changes (e.g., pixel IDs) with highlights
- Tag confidence scores and false-positive feedback
- Slack/Teams integrations for notifications
- Per-Site API keys so agencies can feed Sites in automatically

Final Note

Keep the scope tight. Deliver the smallest useful version, learn from it, then iterate. This document is your checklist—update it as you go.